

EAST AMATULI STATION LIGHT, AK, USA

WMO: 994081

Lat: 58.915N

Lon: 151.952W

Elev: 109

StdP: 14.64

Time Zone: -9.00 (NAK)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	9.7	13.9	3.7	6.7	12.1	6.5	7.7	15.3	59.9	33.1	53.2	33.2	14.8	320	0.916

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	4.9	64.2	56.3	61.1	54.5	58.7	54.3	58.5	61.9	57.2	59.9	56.1	58.1	11.0	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
57.2	70.1	58.7	56.1	67.2	57.4	55.1	64.8	56.4	25.3	62.1	24.5	60.3	23.8	58.1	64.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 48.5	41.8	36.5	DB	6.1	68.9	4.1	3.0	3.1	71.0	0.7	72.8	-1.5	74.5	-4.5	76.7	(4)
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	42.0	33.7	33.2	32.8	37.1	44.1	48.8	52.8	54.7	50.4	43.7	36.9	35.0	
	DBStd	9.63	8.95	7.08	7.16	4.87	3.94	3.73	3.47	3.31	3.19	4.21	6.74	7.15	
	HDD50	3243	507	470	532	387	191	65	8	0	31	194	394	464	
	HDD65	8398	972	890	997	837	647	487	378	321	437	659	844	929	
	CDD50	323	0	0	0	1	9	28	95	145	45	0	0	0	
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0	
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0	
Wind	WSAvg	16.6	19.6	18.7	18.3	17.0	15.7	13.0	13.1	13.5	15.2	16.0	18.6	20.5	
Precipitation	PrecAvg	48.8	5.1	3.9	3.0	2.8	2.4	2.1	2.8	4.2	5.9	5.6	5.0	6.0	
	PrecMax	66.5	12.7	9.0	5.4	6.1	4.5	3.6	4.6	7.6	11.6	12.2	13.1	9.4	
	PrecMin	29.6	0.7	1.1	0.5	0.6	0.8	0.6	1.6	1.2	1.8	2.6	0.9	2.4	
	PrecStd	6.7	2.4	1.7	1.3	1.2	1.1	0.7	0.8	1.6	2.1	2.1	2.9	2.0	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.8	43.1	44.0	53.2	63.9	67.3	66.5	68.4	59.4	51.3	48.3	46.2	
		MCWB	36.4	41.6	41.4	45.0	50.2	59.4	58.4	58.5	52.4	46.0	47.5	45.1	
	2%	DB	44.2	42.2	42.8	45.9	55.3	60.1	63.1	64.6	56.7	50.1	47.1	44.8	
		MCWB	39.5	40.6	40.8	41.8	45.0	51.5	56.4	57.6	53.3	47.0	46.5	43.6	
	5%	DB	43.2	41.4	42.1	43.4	51.7	56.3	60.0	61.6	55.3	49.2	46.1	43.8	
		MCWB	40.7	39.9	40.2	41.0	43.4	50.8	55.1	55.2	53.0	47.2	45.5	42.5	
	10%	DB	42.2	40.6	41.0	41.9	48.9	53.2	57.8	59.2	54.3	48.4	45.0	42.8	
		MCWB	39.8	39.0	39.3	40.0	43.8	49.3	54.6	55.2	52.2	46.6	44.2	41.5	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	43.0	41.8	42.5	45.3	51.2	55.8	60.0	61.1	56.3	50.1	48.3	46.3
			MCDB	43.2	41.9	43.5	52.5	62.9	59.1	64.5	65.1	56.5	50.2	48.3	46.4
2%		WB	41.8	40.8	41.5	43.4	48.0	53.4	58.1	59.1	55.1	49.0	47.1	44.9	
		MCDB	42.4	41.1	42.5	46.1	53.4	55.7	61.6	62.6	55.4	49.2	47.2	45.2	
5%		WB	40.9	39.7	40.5	41.6	46.5	51.5	56.8	57.8	54.3	48.2	46.1	43.7	
		MCDB	41.8	40.2	41.5	42.6	49.3	53.9	59.7	60.8	54.7	48.4	46.4	44.0	
10%		WB	39.9	38.9	39.3	40.7	45.8	50.2	55.6	56.8	53.4	47.4	44.9	42.5	
		MCDB	41.0	39.6	40.3	41.6	48.0	52.1	58.1	59.2	53.9	47.7	45.3	43.1	
Mean Daily Temperature Range	5% DB	MDBR	5.6	5.6	5.4	4.6	4.5	5.1	5.0	4.9	3.8	3.8	5.5	5.8	
		MCDBR	3.9	4.3	4.4	5.4	9.9	10.8	9.2	8.5	5.0	3.4	4.9	4.8	
		MCWBR	4.0	5.0	4.7	3.7	4.6	5.3	4.8	4.4	3.9	4.2	4.6	5.2	
	5% WB	MCDBR	3.6	4.9	4.7	5.2	7.3	8.3	8.3	7.1	3.9	3.6	4.1	4.1	
		MCWBR	4.3	5.4	4.7	3.6	4.3	5.4	5.4	4.9	3.8	4.1	4.4	5.1	
Clear-Sky Solar Irradiance	taub	0.272	0.314	0.327	0.368	0.382	0.368	0.370	0.364	0.347	0.324	0.289	0.280		
	taud	2.269	2.347	2.375	2.298	2.319	2.450	2.491	2.502	2.499	2.394	2.208	2.059		
	Ebn at Noon	177	226	259	265	269	275	272	266	251	218	161	116		
	Edh at Noon	14	21	27	35	36	32	30	28	24	20	14	12		
All-Sky Solar Radiation	RadAvg	108	288	708	1142	1528	1669	1501	1206	807	403	156	69		
	RadStd	15	47	107	145	154	148	145	144	113	40	24	7		

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (12 neighbors)	+1.51	+4.21	+3.92	N/A	N/A	N/A	N/A	-405	-517	N/A	N/A

Nomenclature: See separate page